

Course: T Levels, Digital Production, Design and Development

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Activity Appendix (Research Notes)

These notes contain only facts and figures from research on digital tools in the UK food retail and local produce sector. This Research covers hardware and software usage, emerging technologies in the industry and how digital features support different users and other relevant industry regulations and guidelines.

Hardware and Software in the UK Food Retail and Produce Sector

- Online grocery sales in the UK reached approximately £24 billion in 2025 (IMARC Group, 2025).
- UK online grocery market projected to grow at CAGR of 2.36% from 2026 to 2034, reaching £29.8 billion by 2034 (IMARC Group, 2025).
- Online food and grocery retailing returned to real-terms growth in 2024 after pandemic period (Mintel, 2025).
- Proportion of online food retail sales stabilised around 10% in late 2025 (Statista, 2025).
- Take-home grocery sales grew 3.8% in early 2026 period (IndexBox / Worldpanel by Numerator, 2026).
- Food inflation reached 3.9% in January 2026 (British Retail Consortium, 2026).
- Cloud-based inventory and supply chain tools used widely in food retail (Gartner reports, 2025).
- POS systems and mobile payment hardware standard in most UK supermarkets and convenience stores (industry hardware adoption data, 2025).
- E-commerce platforms adopted by increasing number of small and local food sellers (Statista retail trends, 2025).

Newly Emerging Technologies

- Global food traceability and blockchain solutions market valued at US\$41.56 billion in 2024, projected to reach US\$97.17 billion by 2032 at CAGR 11.2% (DataM Intelligence, 2025).
- Blockchain food traceability market expected to grow from USD 3,037 million in 2025 to USD 52,210 million in 2035 at CAGR 32.9% (Future Market Insights, 2025).

- AI and machine learning used for demand forecasting and waste reduction in food supply chains (IBM and McKinsey reports, 2024–2025).
- IoT sensors deployed for temperature monitoring and freshness in food logistics (IoT Analytics, 2025).
- Combination of AI, blockchain and IoT increasingly applied for farm-to-fork visibility (Food & Drink Network UK, 2025).
- 5G rollout supports faster real-time data in retail and agriculture (Ofcom updates, 2025).
- Smart packaging and digital labels emerging for traceability and consumer information (industry trend reports, 2025–2026).

How Digital Solutions Support Different User Groups

- Convenience remains main driver for online grocery use (Mintel consumer surveys, 2025).
- High percentage of consumers seek transparent origin and sustainability details when buying food (Kantar consumer insights, 2025).
- Accessibility barriers persist on many retail websites; 80% of top UK retail homepages showed critical WCAG issues in 2025 testing (Level Access / Retail Technology Review, 2025).
- Average of 71.2 accessibility errors detected on retail homepages in recent global analysis (WebAIM Million Report, 2025).
- Retail sector identified as having high access barriers for disabled users (UK Government accessibility research, 2025).
- Loyalty and discount features used widely for value-seeking shoppers (Consumer Reports and Mintel, 2024–2025).
- Simple interfaces and guides improve completion rates for new or less confident users (UX studies, 2025).

Industry-Specific Guidelines and Regulations

- Food Standards Agency (FSA) requires accurate labelling and traceability for all food sales including online (FSA guidance, 2025).
- GDPR applies to customer data in digital retail; breaches can lead to fines up to 4% of global turnover (ICO, 2025).
- Equality Act 2010 places duty on service providers to make reasonable adjustments for accessibility (Gov.uk, 2024–2025).

- WCAG 2.1 AA level recommended/required for public and many private sector websites (accessibility standards, 2025).
- PCI DSS mandatory for secure card payment processing in e-commerce (PCI Security Standards Council, 2025).
- Consumer Rights Act 2015 covers accurate product information and cancellation rights for online purchases (Citizens Advice, 2025).
- Advertising Standards Authority (ASA) enforces rules against misleading claims on prices or product benefits (ASA, 2025).
- Green Claims Code requires accurate environmental and sustainability statements (Competition and Markets Authority, 2025).

These notes are limited to raw facts and figures only.

(The search is conducted using AI)

Proposal for the Digital Solution

User Stories

User stories describe what different people need from the new GLH digital platform. They are grouped by main user type (customers, producers, GLH administrators). Each includes a clear role, goal, benefit and testable acceptance criteria. Stories are ordered roughly by importance to core functions (browsing/ordering first).

Customers User Stories:

As a ...	I want to ...	So that ...	Acceptance Criteria
First time visitor	create a customer account quickly	I can start browsing and save my favourite	Should be able to Navigate to signup or login page, should be able to enter email and password etc, get their emails verified and get their accounts active
Registered shopper	sign in securely	I can access personal features like order history etc	Should be able to enter credentials and redirected to dashboard or home
Forgotten password user	recover access through email	I can get entry without help	Can use Forgot password option to receive reset link and set new password

Local food enthusiast	browse products range with photos	I can discover fresh available products easily	Can open products categories and be able to see images, prices and stock status
Specific item seeker	search by keyword or product name	I find exactly what I need fast	Must be able to type in search bar and see relevant results appear and may use filter further if needed
Choosy buyer	apply filters such as price, category or producer	I narrow choices to match my budget or preferences	Be able to select filters and get results quickly
Supporter of ethical farming	read detailed producer background	I can understand farming approach and local impact	Can see producer name on product and other info such as methods, location or sustainability
Value focused shopper	view clear unit pricing like per 100g / kg)	I can compare the cost effectiveness	Have Product card and details page showing unit price alongside total
Time short parent	add items to basket and select collection slot	I collect shopping on way home	Be able to add quantities, choose collection time and see the availability and confirmation
Home delivery user	schedule delivery to my address	I receive items without leaving home	Be able to enter or select address pick any available slot and add any delivery note
Repeat purchaser	see previous orders and reorder items	I save time on regular shopping	Must be able to Go to order history to view past orders and make quick purchases
Account manager	update personal details like address or phone number	My information stays current	Can go to profile section to make and save any changes and receive confirmation messages

Order tracker	check live status of current orders	I know preparation or ready time	Be able to track order details or timeline such as Preparation time and Collection Time etc
Rewards participant	join loyalty programme	I earn points on purchases	Loyalty section → enrol → see initial points balance Be able to join loyalty program and points balance
Points user	apply accumulated points at checkout	I can reduce order total and save money	Must be able to use their points at payment and see the discount applied
Disabled user	activate accessibility options such as large text or contrast	I can use site according to my needs	Have accessibility button such dark, light mode or larger text
Older shopper	use large, clear buttons and simple layout	navigation feels easy	All main buttons must be clear and high contrast
Rural or poor signal user	view basic product info with limited connection	I can still browse when signals are weak	Continuing caching of catalogue pages like PWA features
Quality checker	see harvest or packing date on fresh items	I can choose fresh available produce	Product detail should include dates such as packed on date or harvest date
Feedback contributor	submit star rating and comment after delivery	I can share experience to help others	Have a form which has rating scale and text box to get feedback and submission confirmation
Promotion hunter	view current special offers or bundles	I can take full advantage of deals	Have Home page or banner section which clearly shows active promotions

Social recommender	share product or producer page	I can tell friends or family about good local items	Should have share icons on product or producer pages
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Producers User Stories:

As a ...	I want to ...	So that ...	Acceptance Criteria
Approved producer	log in to secure dashboard	I can manage only my own products and orders	Have producer specific login details to log into their producer dashboard
New supplier	apply to join GLH digital platform	I can start selling through the hub online	Have a registration form where they can submit details and wait for the approval notification
Stock manager	adjust stock quantities in real time	Customers can see accurate availability	Should be able to select products , update number and save, any changes should be visible on the shop instantly
Product updater	edit description, price or photo	Information stays current	Have Edit mode to change fields, preview any changes made and save
Order handler	view and filter incoming orders	I can prepare items quickly	Be able to see orders list sort them by date, status or customers
New item adder	create new product listing	I can offer seasonal or new lines	Have Add form where they can upload photo, enter details of the product
Performance reviewer	access sales and summary charts	I can see which items sell best	Be able to see weekly or monthly graphs or trends showing the sales
Alert recipient	get notifications for low stock	I can restock before running out	get automatic email or dashboard alert when stock below set level

Profile maintainer	update farm or producer description	Customers can learn more about my methods	Be able to edit their profile and save changes and see them on product pages
Feedback reader	see customer reviews on my products	I can improve quality based on comments	Be able to access reviews tab and see average rating and individual comments

GLH Administrators User Stories:

As a ...	I want to ...	So that ...	Acceptance Criteria
GLH staff member	approve new producer applications	Only verified suppliers can join	Be able to access admin panel and see pending list to review approve or reject any producer
Scheme manager	organise loyalty point rules	Can see rewards work fairly	Be able to go to settings adjust earn rate and redemption thresholds and save
Overall monitor	view total hub sales and trends	I can report to cooperative members	Have dashboard to overview and see figures and graphs
Support handler	respond to customer or producer queries	Issues can be resolved quickly	Have messages inbox through which they can reply
Content editor	update homepage banners or news	The site stays fresh and informative	Be able to edit text/images and publish changes in CMS section
Stock overseer	see combined stock levels across producers	I can see any potential shortages	Be able to view or filter by category for low stock alerts
Report generator	export monthly and quarterly data	I can prepare summaries for meetings	Have reports tool, be able to select period and export in file form

Delivery coordinator	Assign and monitor delivery routes	I can see orders reach customers on time	Have Delivery section to be able to see view pending orders and assign driver
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Business Context:

Business Problem:

Greenfield Local Hub (GLH) only uses its one shop building right now. Customers must go to the shop in person or call them up to check what is in the stock, see the details about the food and then place an order. This makes shopping difficult for people who live far away from the shop, work long hours or can't come around easily. Farmers and producers have to write the stock numbers on paper or call the shop to report what they have in the stock which can cause stock arrears or problems for both farmers and customers as they may not get what they want resulting in frustration. This also causes delays as sometimes too much stock is sold by mistake and the food goes to waste if it does not sell quickly on time and everyone spends extra time on paperwork.

GreenField Local Hub (GLH) doesn't have an online shop page that can show clear information about each farmer and how they grow the food, the products that they sell and what they have in stock. Not having an online site is stopping GLH to attract more farmers and customers and grow their business. There is no loyalty scheme to give regular customers discounts. There are no easy access features to help people with disabilities and because of this, GLH only reaches a small part of the local area, misses chances to sell more online and has to compete with big supermarkets that already let people order from their phones.

Key Opportunity:

A new easy-to-use website will give GreenField Local Hub (GLH) a big chance to grow while still supporting local farmers. Customers can look at fresh local food any time of day or night, read about each farmer and their methods, check what is in stock right now, order for collection at the shop or delivery to their home, track where their order is and earn points for shopping again. Producers get their own private page where they can change stock numbers straight away, update product details, see new orders coming in, get warnings when stock is low, and look at simple sales charts. This cuts down the need of manual paperwork and helps them plan better, reduces mistakes and makes the process a lot more fast and efficient. GLH staff can approve new farmers, check overall sales, manage special offers and make sure every item shows exactly where it came from. The site will also have big text options, high contrast colours and dark/light mode to support so everyone in the community can use it easily. The website starts simple with the main shopping and farmer tools, then can grow later with things like special local deals or links to delivery drivers. This will help the business grow attract more

customers and farmers which can lead to people buying locally making it beneficial for everyone and the local economy and also change matches the way more people in the UK now buy food online and will also help GLH build a stronger local food community, cut down on waste with better stock check and bring everyone closer together.

Empathy maps:

To understand the users who will use the GreenField Local Hub (GLH) Website, I've created the empathy maps for producers, GLH staff and customers. Each has four clear parts explaining what they think, what they say, what they do and what they feel.

Producer Empathy Maps:

Thinks:

- How much of this batch is left in stock?
- When will the next orders arrive?
- Does my product description sound good?
- How can I sell more this week?
- Is the farmer page simple to use?
- What if I make a mistake with the numbers?
- Can I see what customers think of my food?
- Is GLH showing my products to more people?
- Can I add new photos of my fields?
- What do I do when stock gets low?
- Are my prices fair compared with others?
- How can I reach more families in the area?

Says:

- I need to change my stock level now.
- Let me see the orders for today.
- I want to update the price on these eggs.
- Add my new apple juice to the list.
- Show me my sales for this month.
- This farmer page is easy to use.

- How do I change the description?
- Send me a message when stock is low.
- Let me read the customer comments.
- Approve my new account please.
- Can you promote my organic vegetables?
- Check how well my products are selling.

Does:

- Signs into the farmer page.
- Changes stock numbers.
- Updates product information.
- Looks at the list of new orders.
- Adds a new item for sale.
- Views the sales charts.
- Reads the low-stock messages.
- Changes farm details.
- Looks at customer reviews.
- Sends in new product photos.
- Checks overall sales numbers.

Greenfield Local Hub (GLH) Staff Empathy Map:

Thinks:

- Which new farmers should I approve?
- Are total sales going up this month?
- How should I set the loyalty rules?
- What do I do about customer questions?
- What news should I put on the home page?
- Is there enough stock across all farms?
- What reports do I need to prepare?
- How do I plan the delivery times?

- Is the website safe from problems?
- Are we following all the rules?
- How can I tell people about new offers?
- How many people are using the site?

Says:

- Approve this new farmer.
- Show me the total sales figure.
- Change the loyalty discount level.
- Reply to this customer message.
- Add the new farm story to the page.
- Check all the stock levels together.
- Create the monthly sales report.
- Assign this delivery to the driver.
- Run a quick security check.
- Make sure we follow the rules.
- Send out the weekly newsletter.
- Look at how many visitors we have.

Does:

- Signs into the staff control page.
- Checks and approves new farmers.
- Looks at overall sales numbers.
- Changes loyalty settings.
- Answers customer messages.
- Updates the home page.
- Checks combined stock across farms.
- Makes and saves reports.
- Organises delivery schedules.
- Checks the rules are followed.

- Sets up new special offers.
- Reviews visitor numbers.

Feels:

- In charge of the whole system.
- Busy keeping everything running.
- Happy when sales are growing.
- Careful about any problems.
- Quick and organised with the tools.
- Proud to support the cooperative.
- Pressured when the site is very busy.
- Calm when reports look good.
- Sure the website is working well.
- Keen to help customers and farmers.
- Responsible for the success of GLH.
- Working well as part of the team.

Customers Empathy Map

Thinks:

- Is this food really fresh from a local farm?
- Is the price good value for money?
- Can I get my shopping delivered this afternoon?
- Is my personal information safe on the site?
- Why should I choose local food over the big supermarket?
- How many points have I earned for my next shop?
- Will the website work well on my phone?
- What happens if my order is late?
- Does this product have any allergens I need to know about?
- How do I check where my delivery is right now?
- Can I compare different vegetables easily?

- Will the delivery van use less fuel than driving myself?

Says:

- Where can I find the organic carrots?
- Tell me more about this farmer's farm.
- Can I change my delivery time to after school?
- The loyalty points are really useful!
- The page is taking too long to load.
- I like seeing exactly where the food comes from.
- How do I cancel this order?
- Can the site read out loud for me?
- These filters are good but I need more choices.
- I forgot my password again.
- Show me the special offers on milk and cheese.
- Is this food grown without chemicals?

Does:

- Looks through the product pages.
- Puts items into the shopping basket.
- Signs into their account.
- Chooses a collection or delivery time.
- Checks past orders.
- Changes their address or phone number.
- Uses loyalty points at checkout.
- Searches and picks filters.
- Writes a short review after getting the order.
- Sends a message to customer help.
- Shares a product link with family.
- Looks at the farm information.

Feels:

- Happy to support local farms.
- Annoyed if the site is slow.
- Pleased when ordering is quick and easy.
- Safe knowing the information is clear.
- Glad to save money with points.
- Worried about delivery times.
- Calm when the tracking works.
- Sure the payment is safe.
- Thankful for big text and easy buttons.
- Upset if favourite items are out of stock.
- In control of what they buy.
- Part of the local community.

These mapss help show what users think, say, do and feel. This will make the website better for everyone.

Functional Requirements :

The system must let new customers sign up with their name, email, password, address and agree to the rules before they can shop.

The system must let customers, producers and GLH staff log in safely with their email and password.

The system must let users change their forgotten password by sending a link to their email.

The system must let customers change their address, phone number or other details and see all their past orders in one place.

The system must let new producers apply to join and wait for GLH staff to check and approve them.

The system must let every user log out safely so no one else can use their account.

The system must show a clear home page with pictures of fresh products, latest news and easy buttons to click.

The system must have a full shop page where customers can look at every product with photos.

The system must let customers see all the details about each product, including the price, how much is left in stock, who grew it and where it came from.

The system must let customers search for any product by typing a word and use buttons to sort by price or type of food.

The system must show special offers and money-saving deals on the home page and shop page.

The system must let producers add new products with a name, price, short description and a photo.

The system must let producers change any product details, price or photo whenever they need to.

The system must let customers put products into a basket and go to the checkout page.

The system must let customers choose collection at the shop or delivery to their home and pick a time that suits them.

The system must send an email to the customer when the order is placed and another message to the producer so they know what to prepare.

The system must let customers see where their order is right now with simple steps like “being packed” or “ready to collect”.

The system must take payments safely using a trusted card service. The system must show customers all their old orders with dates and what they bought.

The system must let customers join the loyalty scheme and earn points every time they buy something.

The system must let customers use their points to get money off at the checkout.

The system must show the customer how many points they have at any time.

The system must give producers their own private page where they can log in and work.

The system must let producers change stock numbers straight away so the shop page always shows the correct amount left.

The system must let producers see a list of all the orders for their products.

The system must send producers a quick message when their stock is getting low or when a new order comes in.

The system must give GLH staff a special control page to look after everything.

The system must let staff approve new producers before they can sell on the site.

The system must let staff create simple reports about sales, stock and customers.

The system must let staff change the home page news or add new offers.

The system must let customers write a short review and give stars after they get their order.

The system must check every piece of information people type in so mistakes do not happen.

The system must show friendly messages if something goes wrong instead of confusing errors.

The system must work on computers, tablets and phones so everyone can use it.

The system must use simple frontend code for the pages customers see and strong backend code to keep everything safe and fast.

The system must save all information in a secure database.

The system must let many people use it at the same time without slowing down or crashing.

Non-Functional Requirements:

All customer and producer information must be kept private and safe at all times.

The website must use a secure connection on every page so no one can steal data.

Payment details must be handled by a trusted company that follows the highest safety rules.

Only the right people must be able to see certain parts of the site, for example producers cannot see other peoples orders.

Every change on the site must be recorded so GLH can check what happened if needed.

Every page must open quickly in less than two seconds on normal home internet.

When someone searches for a product the results must appear in less than one second.

The site must still work well even if hundreds of people are shopping at the same time.

Stock changes must show straight away on the customer pages.

The whole website must follow the latest rules such as WCAG 2.1 guidelines so people with disabilities can use it easily.

The site must read out loud properly for screen readers and let people move around with just the keyboard.

All buttons and text must be easy to see with good colour contrast.

All pictures should have a short description that screen readers can use.

The site must stay working almost all the time with copies saved every day in case something goes wrong.

Information must never get lost when it is saved or changed.

If something breaks the site must show a helpful message and not stop completely.

The site must be able to grow if more customers or producers join later.

Cloud servers should be used that can handle busy days like weekends without slowing down.

The pages must be made in separate pieces so new features can be added easily later.

The site must be simple and user friendly even for people who do not use computers much.

Every page layout should look and work the same way so users do not get confused.

The site should change the size automatically to fit users phones and tablets perfectly.

Any error messages displayed should use easy words that anyone can understand.

The site should work on the main browsers that mostly people use in the UK.

The code must be written clearly so other people can understand and fix it if needed.

No customer information must ever be shared without their clear permission.

The site must use clear British English and be ready if other languages are added later.

Open source tools must be used where possible to keep costs low for GLH.

The code must be written efficiently so it does not use too much electricity on the servers.

Key Performance Indicators (KPIs):

Key Performance Indicators (KPIs) are just simple numbers that tells how well the Greenfield Local Hub (GLH) website is performing.

Numbers of customers creating an acciount eeach month.

Counting how many producers lohg into their dashboard and update stock every week.

Check the percentage of customers who join the loyalty scheme.

Number of orders placed on the website each week.

Number of customers come back and order again within three months.

Comparing the total sales each month and comparing to the last month to see the growth.

How long it takes for each page to load on a normal phone connection.

How often the website is working and available all day.

How many errors happen for every 500 people who use the website.

How accurate the stock numbers are by comparing the website with the real stock in the shop.

How many Orders are collected and delivered on time the customer choose.

Number of people who leave the shopping basket without finishing the order.

Average rating customer give after they receive their order.

How many product views are left after each month.

Number of people who use accessibility tools such as big text or light/dark mode.

Number of customers who use the tracking page each week.

Number of minute average customer spends on the website during one visit.

Number of producers who use the sales report tool each month.

All these will help GLH see whether the new website is helping them or not.

System Hierarchy

GLH Digital Platform

User Interface (Front-end):

- Account pages (sign up, sign in, my details)
- Home page
- Product categories page
- Basket and checkout pages
- Order tracking page
- Producer story and info pages

Producer Tools (private area for farmers):

- Producer dashboard
- Stock update page
- Sales report page

Backend (Server Logic Module):

- User Authentication System
- Notification System
- Order Processing System
- Product Management System
- Loyalty System

GLH Staff Tools (private area for hub managers):

- Staff control page (dashboard)
- Reports and approval page

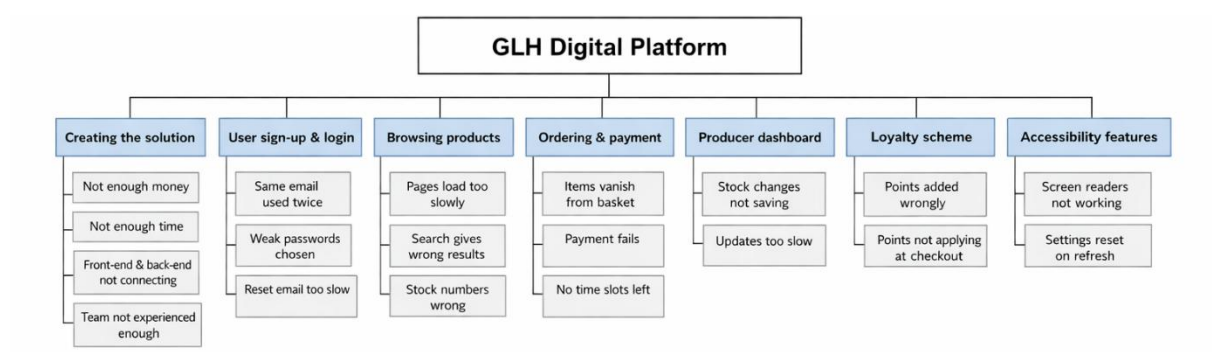
Security Layer:

- Login and password system
- Permission checks

Data Storage:

- Customer records
- Loyalty points records
- Stock records
- Order records
- Product records

Hierarchy Diagram:



Decomposition of the Problem:

The digital solution can be broken down into smaller and clear pieces, so it is easier to build, test and fix if something goes wrong.

Creating the Solution

- Running out of time to test everything properly could leave hidden mistakes in the live website. It can lead to bugs that can stop orders going through, wrong stock numbers showing, customers getting frustrated, lost sales and bad reviews right after launch.

- Not having enough money to pay for extra tools or extra help could mean some important parts of the website are left out completely or finished in a hurry without proper care or coding which can lead to a weaker system that misses key features customers and producers need also leading to extra work to fix things
- Choosing coding tools or languages that you are not confident with could slow down every step of building which can lead to delays and more mistakes in the code and making website less reliable
- The front end which users see and the hidden server parts not interacting to each other correctly could stop important actions like adding items to the basket, updating stock or completing payments. It will lead to broken features causing users giving up on the site and damaging the reputation.

Sign-up and Login Risks

- The email with the password reset link taking too long to arrive or never arriving could make new customers or people who forget their password give up completely. This may lead to fewer sign-ups causing to lose potential shoppers.
- Customers choosing very simple or easy-to-guess passwords: This could let someone else guess the password and get into the account. It can lead to stolen personal information, fake orders being placed, money taken from linked payment methods, angry customers, and loss of trust in GLH.
- Two different people trying to use the exact same email address could mix up accounts. It can lead to confusion, privacy complaints and security weaknesses.

Shopping and Browsing Risks:

- The shop page taking a long time to load or open makes users close the website before they even look at products. This can lead to very few visitors staying long enough to buy anything and creates first and first bad impression.
- The search tool giving wrong results, missing items or showing nothing, useful means customers cannot find the food they want even when it is in stock. This may lead the customers to be annoyed and going to other supermarkets website.
- Stock numbers shown on the website not matching what is available in the shop or from producers could let customers order items that have already run out. This may lead to customers loss of trust, bad reviews can lead to customers being angry when they arrive to collect they get nothing.

Ordering Risks:

- Items disappearing from the shopping basket before the customer finishes could make people very annoyed and stop them from buying.

- Payment not going through even when the card details are correct could stop the order at the last step leading to loss in sales.
- All delivery or collection time slots already taken or showing as full could stop customers from completing their order meaning the users will get disappointed leading to lower customer satisfaction.

Loyalty Scheme Risks

- Points being added or taken away incorrectly like some customers may get too many points and others might get few. This can also cause loss of trust in the loyalty scheme and damage to the reputation of GLH.
- If loyalty points not working properly when a customer tries to use them at checkout, it could stop the discount from appearing even though the customer has enough points leading to customers loss of interest.

Producer Dashboard Problems:

- Changes to stock levels not saving correctly or disappearing means the website shows the wrong amount left to customers. This might lead to overselling, angry customers and extra refunds producers losing money and time and damage to the relationship between producers and GLH.
- Stock updates taking too long to appear on the customer shop pages could let people add items to their basket that the producer has just sold out. It can lead to cancelled orders, unhappy customers and producers, refunds and loss of confidence in the system.
- Someone who is not a producer getting into the producer area by mistake or on purpose could let them see or change stock, prices or orders that do not belong to them leading to have wrong information on the site causing trust problems and security complaints.

Rule and Safety Risks

- Personal information like names, addresses and emails getting stolen or leaked. This could expose customer details to criminals or hackers leading to identity theft, unwanted messages or scams causing customers angry and upset and big fines from data protection authorities and serious damage to GLH's name.
- Wrong or missing information shown about a product such wrong price, wrong origin or missing allergens could mislead customers or put their health at risk. It can further lead to complaints, returns, refunds and possible food safety rule breaks causing GLH fines from authorities.

Accessibility Risks

- Some parts of the site such as buttons, text or images not working properly with screen readers or other help tools could make the website very hard or impossible for people with disabilities to use. It can cause possible legal problems under equality rules damaging the reputation.
- Accessibility settings like big text, high contrast or light dark mode disappearing every time the page refreshes or moves to a new page could force users to keep turning the settings back on again and again. It can lead to customers frustration.

User Acceptance Criteria:

General Checks for the Whole Website

- The website must open without any error messages on the main browsers people use such as Chrome, Firefox or Edge etc.
- All menu buttons and links must take the user to the correct page and show which page they are on.
- Every link in the bottom section (footer) must go to the right place such as About, Contact or Privacy etc.
- The colours, letter sizes and button styles must look the same on every page.
- The accessibility button must appear on every page and keep the chosen settings like big text, high contrast or light/dark mode even when moving to a new page.
- When someone types wrong information such as wrong email format or empty fields the site must show a clear message explaining what is wrong.

Home Page:

- The home page must show a big welcome picture or message about buying local food and some featured products.
- It must load fast and look good on a phone without needing to scroll sideways.
- The main buttons Product Categories, Producers, My Account, Basket etc and others must work and take the user straight to the right place.

Product Shop Page:

- All products must appear with clear photos, names, prices and a note saying if they are in stock or low.
- When a customer types a word in the search box the right products must appear quickly.

- When a customer picks a filter such as price low to high or only vegetables the list must update straight away to show only matching items.
- Clicking on any product picture or name must open the full product detail page.

Product Detail Page:

- The page must show a large photo of the item, the name, current price, how much is left in stock, full description and the name of the producer.
- The “Add to Basket” button must add the right number of items and update the small basket number at the top of the screen.
- The producer information section must show simple facts about the farm and how the food is grown.
- If the product has allergens like nuts or milk etc this must be shown clearly near the top.

Account and Login:

- The sign up form should check that the email looks correct and the password is strong enough before creating the account.
- When a customer logs in with their right email and password they should go straight to their account or home page.
- The account page should let the customer change their address, phone number or password and show a list of everything they have ordered before.
- When someone clicks “Forgot password” they must receive an email with a link that lets them set a new password safely.

Ordering and Basket Page:

- The basket must show every item added, the quantity, the price for each and the total cost so far.
- At checkout the site should let the customer choose collection at the hub or delivery to their home and pick from available times.
- After paying the customer should receive an email saying the order is received with all the details.
- The order tracking page must show simple steps such as received, being prepared, ready to collect or on its way that update as things happen.

- A normal customer must be able to finish an order from adding items to payment in less than two minutes.

Loyalty Scheme Checks

- There must be an easy button for customers to join the loyalty scheme from their account page.
- After a purchase the correct number of points must appear in the customers account straight away.
- At checkout the site should show how many points the customer has and let them use some to get money off the total.
- The points balance should always show the right number in the account and at the top of the screen.

Producer Dashboard:

- Only a logged-in producer should be able to see and use the producer dashboard and no one else.
- When a producer changes the stock number for a product the change should always show immediately on the customer shop page.
- When a producer edits a product such as its price, description or photo the new details should be saved correctly and appear on the site.
- The orders list should show all new orders for that producers items with customer name, items and time needed.

GLH Staff (Admin):

- Only GLH staff must be able to log in to the admin area and see all control options.
- When staff approve a new producer, the producer must get an email and be able to log in straight away.
- The sales report must show correct total money taken, number of orders and top selling items for any date range chosen.
- When staff add or change news or offers on the home page the updates should appear for customers right away.

Performance and Safety:

- All accessibility tools like big text, screen reader mode or light and dark mode should work correctly on every page.

- The site should follow the main rules for keeping customer information safe as described by the GDPR and showing correct food details.
- The website should work efficiently when 500 people are using it at the same time.

Hardwar, Software, Programming language and tools to be used:

For Hardwar a good computer with good specifications will be used o be able to program the solution well.

The software that will be used will be:

- Visual Studio Code – To program the solution.
- Figma – To make flowcharts, diagrams and more for the solution.

Programming Languages:

- React – To make the interactive frontend of the solution
- Java Script (Node.js) – To make and build the backend of the solution.
- SQLite: To make the database for the system to retrieve and save information from

Web development:

- HTML – To create the pages for the solution
- CSS – To create designs and themes for the HTML pages.

Proposed Solution:

The proposed solution will be a safe and easy to use website that brings together local customers and the farmers and food makers who supply Greenfield Local Hub (GLH). It will be a single online place where people can see fresh local food, learn about the producers, place orders and manage their shopping, while the producers can look after their stock and see what customers want. The website will be built using modern web tools and languages. React for the parts customers see and click on so the pages feel quick and smooth and a strong back end system using JavaScript to keep all the information secure, handle all the logins, process orders and send messages. A reliable database SQLite will store everything safely like customer accounts, product details, orders, stock levels, loyalty points and traceability information.

The website will have a clean and consistent look on every page. There will always be a navigation bar across the top with simple buttons such as Home, Product Categories, Producers, My Account, Basket and an accessibility icon etc. At the bottom of every page there will be a footer with useful links like About GLH, Contact, Privacy Policy,

Terms and Conditions etc and copyright information. The accessibility button will allow to choose larger text, light and dark mode or screen reader friendly mode or other help options. These choices will stay saved so the site remembers them next time the person visits.

The home page will welcome visitors with a large photo of local fields or fresh produce and a short message. It will show a few featured products that are in season or on special offer, a search bar, Product Categories and producers and a small section with the latest news or events from GLH.

The Product Categories page will let customers browse all available products in a clear list. Each item will show a good photo, the name, current price, how much is left in stock and a quick note about the producer. Customers can use a search box to find something specific or pick filters such as only vegetables, only organic, price low to high or by producer name so they see exactly what they want.

When someone clicks a product they go to a detail page. This page will have a big photo, the full price, stock level, a clear description of the food, allergen information if needed, the harvest or packing date, and a section called Meet the Producer that tells the story of the farm how the food is grown and why buying it helps the local area and the environment.

Customers can add any product to their basket with one click. The small basket icon at the top will always show how many items are inside and the total cost so far. On the basket page they can change quantities or remove things. At checkout they choose whether to collect from the GLH hub or have it delivered to their home, then pick a time from the available slots shown on a simple calendar. They can use any loyalty points they have earned to get money off, then pay safely using a trusted payment service. After payment they get an email confirming the order and they can track it on their account page to see simple updates like being prepared, ready to collect or on its way.

Every customer gets their own account area after they sign up or log in. Here they can see all their past orders, check current order status, change their delivery address or phone number, view how many loyalty points they have and see any special offers just for them. They can also join the loyalty scheme if they have not already earning points on every purchase and using them for discounts.

Producers the farmers and makers will have their own private dashboard that only they can reach after logging in. On this page they can quickly change how much stock they have left, edit product details like price, description, photos, see a list of all their incoming orders with customer names and what is needed, get a warning message when stock is running low and look at simple charts showing which of their items are selling best.

GLH staff will have a separate secure area where they can approve new producers who apply to join, check overall sales and stock across all suppliers, create reports to share with the cooperative, update news or special offers on the home page and make sure everything follows the rules for food safety and data protection.

The whole website will work perfectly on phones, tablets and computers changing size automatically so it is always easy to read and use. It will include strong safety features such as secure logins, protected payments and full compliance with data protection laws. In the future, extra things like better traceability records using modern methods could be added but the first version will focus on the core needs like easy shopping for customers, simple stock and order tools for producers, and clear management for GLH.

This solution directly solves the problems in the task brief of the client and investors. It gives customers convenient online access to local food with full information about producers and traceability, lets them order for collection or delivery, manage accounts and use loyalty rewards. It gives producers a straightforward way to update stock, see orders and manage products. It helps GLH promote local food better, keep accurate records, improve efficiency and reach more people in the community.

Justification of how the recommended solution meets the needs of the client and users:

The proposed website will fully meet the needs of Greenfield Local Hub (GLH), the customers and the producers in a simple and practical way. It will directly solve the problems described in the task brief and will turn the opportunities into real benefits for everyone involved.

For GLH the website will strongly support the main goals of the cooperative. It will promote the local food by putting every producers story, farming methods and traceability information online so more people in the community can see and trust where their food comes from. It will improve day to day running by moving stock updates and order information from paper and phone calls to a digital system that is fast and accurate. This will reduce mistakes, save time for GLH staff and will help avoid waste when products do not sell quickly.

The website will open up new ways to earn money through online orders that customers can place any time, day or night and through the loyalty scheme that encourages people to shop again and again. Because the site will be built with strong security, secure payments and clear rules for data and food information, it will follow important laws such as GDPR for protecting personal details and Food Standards Agency rules for accurate labelling and traceability. This keeps GLH safe from legal trouble and builds trust with the local area.

For customers the website will make buying local food much more convenient and enjoyable. They no longer will need to travel to the hub or phone to find out what is available and what is not. They will be able to see and browse all full range from home or on their phone at any time. Every product will show clear prices, current stock levels and all details about the producer and how the food is grown so customers will know exactly what they are buying and why it is good to choose local. Ordering will also be easy, they customers will be able to add items to a basket, pick collection at the hub or delivery to their door, choose a time that suits them, use loyalty points for discounts, pay safely and get an email straight away. They will be able to track their order with simple updates like they will see all their past orders in their account and manage their details whenever needed.

The accessibility features will make the site usable for older people and those with disabilities or anyone who finds small text or complicated menus difficult. This means more people in the community can enjoy fresh local food without barriers.

For producers such as the farmers and food makers, the website will give them simple digital tools that will make their work easier and will help them sell more. They would have their own private dashboard where they can log in and change stock numbers in seconds so the customer can see the true amount left leading to no more phone calls or paper notes. They will be able to update product details, add new items with photos and descriptions, see every order that comes in for their produce with customer names and what is needed and receive quick alerts when stock is getting low. This will save them time reducing the risk of selling things they do not have and also lets them plan growing and harvesting better. Because the website shows clear information about their farm and methods their products will get more visibility and customers will feel good about choosing them. This can lead to more regular orders and stronger connections with local buyers.

The website will use straightforward and reliable technology that is powerful enough to do everything needed without being complicated or expensive to run. The front end will be built to feel quick and friendly on phones and computers, the back end will keep all information safe and up to date and the database holds everything in an organised way.

By solving the current problems of limited reach, slow manual processes and lack of efficiency, the solution will bring value to the cooperative helping local food sell better and will support the producers and makes shopping enjoyable and trustworthy for customers.

Justification of how potential risks will be mitigated:

Risks during building the website

Not enough money for tools or extra help can leave features unfinished or rushed. - Planning the budget early and use free open source tools.

Not enough time to test everything can leave bugs in the live site - Make a clear timetable and test the most important parts first.

Front end pages and back end server not connecting properly can break ordering and stock updates – Testing connections between front end and back end at every step.

Risks with sign-up and login:

Two people using the same email can mix up accounts and details - Check every email is unique before saving the account.

Weak passwords can let someone break into accounts - Force strong passwords with letters, numbers and length rules.

Password reset email arriving late can make users give up signing up - Use a fast, reliable email service to send links quickly.

Login details not saved correctly can stop users getting in - Test saving and fetching accounts many times before launch.

Risks with browsing products:

Pages loading too slowly can make customers leave quickly - Make images smaller and load only a few products at a time.

Search giving wrong or no results can stop customers finding food - Have a clear search that checks names, descriptions and categories.

Stock numbers not matching real stock can let people order sold out items - Update stock numbers quickly when producers change them.

Risks with ordering and payment:

Items disappearing from basket can annoy customers and stop buying - Save the basket on the device until checkout is finished.

Payment failing when details are correct can lose sales - Offer more than one safe payment method and test each one.

No time slots left for delivery or collection can block orders - Lock each time slot when a customer chooses it.

Risks with producer dashboard:

Stock changes not saving correctly can show wrong amounts to customers – Double check every save before it is finished.

Stock updates arriving too slowly can cause overselling - Send new stock numbers straight to customer pages in real time.

Wrong person getting into producer area can change things they should not - Check login role every time the dashboard is opened.

Risks with loyalty scheme:

Points added or taken wrongly can create unfair discounts - Test the points rule many times before launch.

Points not applying at checkout can stop discounts - Check points balance is correct before showing the discount.

Risks with accessibility features:

Some parts not working with screen readers can leave disabled users out - Check every page against accessibility rules from the start.

Accessibility settings resetting on page change can frustrate user - Save chosen settings so they stay for the next visit.

Risks with rules and safety:

Personal information getting stolen can cause identity theft and scams - Use strong encryption and secure connections everywhere.

Wrong product information shown can mislead customers or risk health - Let GLH staff check every new product before it goes live.

Legal and Regulatory Considerations:

The website will follow all important UK laws and rules for an online food shop run by a cooperative like Greenfield Local Hub. This will keep everything safe, legal and fair for customers, producers and GLH.

Customer personal details like their names, address, email and payment information will be protected under GDPR rules. Data will only be collected when really needed for orders, delivery and loyalty points. It will be stored with strong encryption, only staff who need it will be able to see it and customers will see a clear privacy notice or a consent box when they sign up or buy something. They will be easily be able to view, change their data or delete their data from their account page and no data will be shared without permission and will be kept securely.

All product information will meet consumer protection laws and Food Standards Agency (FSA) food safety rules. Every product page will show true prices, exact stock levels, full descriptions, producer name, origin, harvest or packing date and clear allergen warnings. Customers will get a full order confirmation email right away.

The website will follow the Equality Act 2010 and WCAG 2.1 AA accessibility standards so everyone can use and access it easily. An accessibility button will stay on every page. Users can choose larger text, choose light or dark mode, screen reader support and keyboard only navigation. All the images will have clear descriptions of products and buttons that will be easy to tap. These features help older people, people with disabilities and anyone who finds small text difficult.

Payments will be handled securely under PCI DSS rules. No card details will be kept on the GLH website. All payments will go through a proper certified payment company using secure HTTPS connections on every page.

Advertising and the claims made will follow Advertising Standards Authority (ASA) rules. All prices, discounts, local or fresh statements and special offers will be honest and clear. Nothing misleading will be shown about products, origins or benefits.